

Pilot-Proxy Findings Ledger

A living supplement of dissertation-grade results behind the RASTI article

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Purpose. The RASTI article keeps only what its argument needs. This ledger keeps the rest: measured findings, methodological lessons, and negative results that belong in the dissertation’s narrative even where the article compresses them to a clause. Every number here maps to a command in the public repository or an archived provenance memo; sections cite the source. The full supplementary plot set (all-23 spectra in three variants, per-channel F distributions, diurnal and stratified diagnostics) travels separately as `supplemental_plots_rev16.zip`.

1 ch30 dissected: transmitter silences, not fades

The refused channel ch30 is cleanly bimodal (Fig. 1): 96.3 per cent of frames sit in a tight clump at $11.7 \times \mu_0$, the range $1.5\text{--}10 \times \mu_0$ is essentially empty (0.26 per cent), and a 3.46 per cent minority (314 frames) sits below. The minority is 38 *whole* captures — not one capture mixes the two populations — confined to 2019 April 6–25 plus one day in October. That structure rules out propagation fades: the dominant occupant simply went off the air, twice, in 2019, and never again in 7.6 years of sampled exposure. The census makes the attribution nearly forced: CHKL-1 (Relay, Penticton BC) sits 17.7 km from DRAO at 79.3 dB detectability — 25 dB clear of the runner-up (CH2520-DT, 156 km) — and a 17.7 km path never fades into the gap, which is why ch30’s mask is effectively deterministic presence detection. One nuance worth keeping straight: in our census source the *Relay* class carries a ± 1 kHz tolerance entry; it is the *translator/LPTV* class that records none (the ch33 story, §8). ch30’s measured drift picket across its guard band is therefore evidence of poor carrier discipline in practice, not of a permissive rule on paper. The off-air interval is a datable, checkable event (ISED records; pending).

Two more results fall out of the dissection. First, the minority’s centre lies within 1.5×10^{-3} of the *analytic* zero point: the moment the transmitter goes silent, the channel relaxes to exactly where the detector model says a clean channel sits — direct, per-channel evidence that the refusal is transmitter-caused rather than instrumental. Second, the minority is *not* a usable calibration core: within-capture width 37×10^{-3} (fifteen times the trusted-channel null width), capture means scattered 18×10^{-3} rms in both directions (weak residual co-channel energy in target *and* reference cells once the dominant is off), occupancy 0.5 per cent against the 15 per cent floor. The mixture-model reclamation route, applied to the one channel that invites it, confirms the refusal. Repro: `analysis/ch30_offair_minority.py`; `ch30_offair_minority.csv`.

The ch24 contrast. The archive’s other refusal dies by the opposite geometry. ch24 has *no* shape to cut: its core is displaced $+20.4 \times 10^{-3}$ (signal-elevated by persistent in-cell carriers), occupancy 7.6 per cent, a continuum rather than a bimodal split — there is no gap for a mixture model to exploit. And no more samples are coming: ch24 is already sampled at full archive depth (1722 of 1722 events), so the running survey extension adds nothing there. Only new observations could change its picture, and the persistent-occupant physics says they would not. The pair makes a tidy taxonomy: **ch30 is separable but its null is unusable; ch24 is unseparable outright.** Both refusals are final on this archive.

The time dissection (Fig. 2) completes the contrast. ch24’s F never leaves the band $\sim 0.93\text{--}1.3 \times \mu_0$ (nothing above 1.3 beyond 0.2 per cent; compare ch30’s clump at 11.7), and it is *two-sided*: 26 per cent of frames sit below $0.99 \times \mu_0$, more than 4σ under the null — measured proof that co-channel energy lands in the *reference* cells and pushes F down, the same mechanism seen on ch30’s off-air wings and the empirical leg

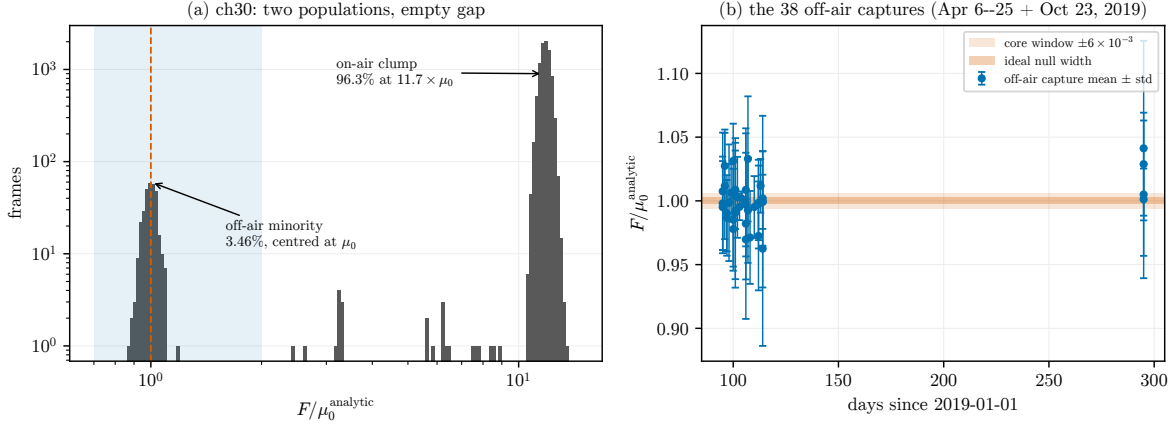


Figure 1: ch30's two populations. (a) The on-air clump at $11.7 \times \mu_0$ and the off-air minority at μ_0 , with an essentially empty gap between. (b) The 38 off-air captures: means within a few 10^{-3} of the analytic zero point, but widths far beyond a true null core.

of the purity-is-not-monotonic argument (§6). Against time it is a stationary committee: quarterly medians hold at 1.02–1.05 from 2018 Q4 through 2026 Q2 with no step, no repack signature, and no silence — 4 isolated quiet captures out of 1722 (0.2 per cent, scattered single days: momentary deep fades, never a window), versus ch30's 38-capture three-week silence. The census explains the difference in kind: ch30 is a monolith (one 17.7km relay, 25 dB clear), while ch24's list is led by the CHKL-DT+CHBC-DT channel-share at 62.8km with relay and translator members behind it — a committee is never all off at once. One genealogical note worth keeping: CHKL-DT, the ch24 share parent in Kelowna, is the parent station of CHKL-1, the ch30 Penticton relay — the same broadcast family plausibly occupies both refused channels (census candidates; ISED corroboration pending), and the parent shows no interruption during the relay's 2019 April silence. Sampled-epoch caveat: the ch24 archive itself has coverage holes (2021–2024 Q2 and 2025 hold no captures) — an archive property, not a transmitter property; within every sampled epoch the committee is stationary.

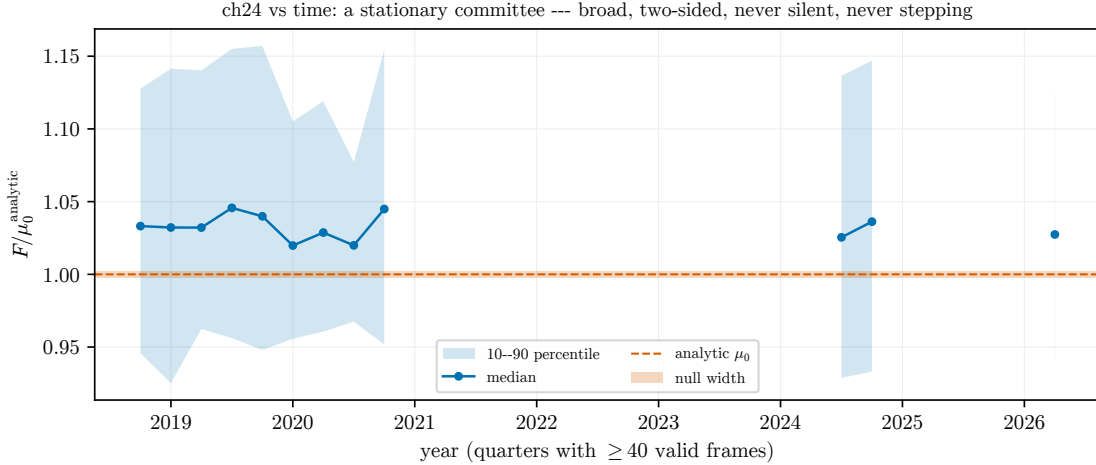


Figure 2: ch24 against time: quarterly median and 10–90 percentile band of F/μ_0 . Broad, two-sided about a displaced core, stationary across 7.6 years, never silent — the committee counterpart to ch30's monolith.

2 The T2 saga: a generator startup transient

The synthetic golden capture’s pilot read ~ 0.5 dB low against its own specification, and the deficit *moved* (0.53–0.72 dB) depending on which estimator and sample span measured it. The resolution came from a ten-block coherent projection (Fig. 3): block 0 sits -9.71 dB and block 1 -0.41 dB below the stationary mean of blocks 2–9 — the waveform generator takes ~ 11 ms to reach steady state, and every full-span scalar inherited a diluted average of that ramp. Once cropped to the stationary span and retrimmed ($\times 0.98447$), two independent estimators agreed to 0.007 dB, and the steady-state pilot sits 0.136 dB *above* nominal. Lessons worth keeping: (i) a “span-dependent scalar” is a symptom of non-stationarity, not of a broken estimator; (ii) the two estimators’ 0.194 dB convention gap was itself stable through the trim — measured, predicted, then closed; (iii) always discard the first ~ 120 k samples at generation. Provenance: `data/provenance/t2.convention.20260718/`; T2_RESOLUTION and R9/R10 triage memos.

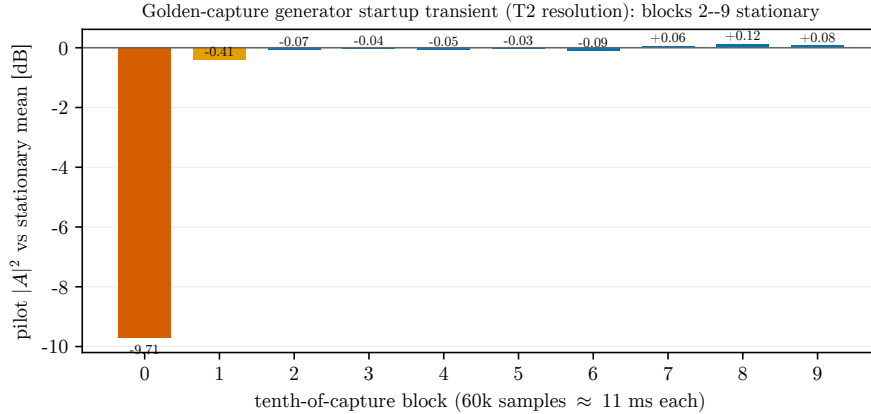


Figure 3: The T2 resolution in one picture: the golden capture’s pilot power by tenth-of-file block. The deficit was never in the detector — it was the generator warming up.

3 A falsified physics alternative

The one mechanistic model we built for the zero-point shifts — the aliased-noise gain of the 4-tap sinc-Hamming reference PFB prototype through the $K=128$ window — *fails*: it predicts -1.8 to -4.2×10^{-3} for the near-edge channels 18/29/32/35, where the measurement is consistent with zero, and captures only a third of ch21’s -12×10^{-3} wrap shift (`fig_pfb_zero_point_prediction`). The deployed F-engine coefficients evidently differ from the reference prototype. The dissertation-grade point: we kept the falsification in the paper and let the measured 23-entry table be the calibration — “the measured table, not a mechanism” is an honest and defensible calibration philosophy when the deployed instrument diverges from its documentation.

4 From magic numbers to a derived criterion

The null-core trust flags began life as placeholders (0.9 mask fraction, 15 per cent occupancy, 20×10^{-3} range). They ended as one derived criterion measured two ways plus a geometry bound: because the positive-excess rule masks null frames at 0.5 by construction, a mask fraction > 0.9 *is* the statement $f_{\text{null}} = 2(1 - \text{mf}) < 0.2$; the core-window occupancy floor is the same f_{null} by direct count (the $\pm 6 \times 10^{-3}$ window is 2.5 ideal null widths); and the displacement bound 20×10^{-3} is $1.7\times$ the largest geometry-producible shift (ch21’s wrap). The archive then shows the constants are not delicate: trusted channels sit at occupancy ≥ 19.5 per cent and $|\text{gap}| \leq 12 \times 10^{-3}$; refused ones at ≤ 7.6 per cent and $\geq 20.4 \times 10^{-3}$. Bimodal separation is what makes a threshold criterion robust — the same structural fact, twice, at two scales of the analysis (cf. §1).

5 The width-crossing scale and its universal curve

The detector’s sensitivity has a natural unit: the shelf SNR s at which the induced mean shift Cs equals the null width $\sigma = \sqrt{1/R + 1/2R}$. In the reduced variable $k = Cs/\sigma$, the detection probability collapses onto one curve across dimensionalities — $P_d(k=1) = 0.833$ at -34.30 dB for 512-row trials and 0.841 at -47.85 dB for full frames — so “per-trial” and “deployment-scale” quotes are the same physics at different R . The flip side is a universal caveat the tiering must carry: signal below the width crossing retains at ≈ 0.5 on *every* channel by construction. Sensitivity claims and retention claims are two readings of one number.

6 Why the threshold sits at the mean, not deeper in the left tail

The natural challenge to $\tau = \hat{\mu}_0$ is: if kept data must be clean, why not keep only the far left tail? The quantitative answer (Fig. 4) is that the null and a weak-signal population are the *same distribution shifted by $k\sigma$* , so a deeper cut buys purity only at the Gaussian hazard exchange rate — and pays for it exponentially in exposure. Moving the threshold from the mean to 2σ below it keeps 2.3 per cent of clean frames instead of 50 per cent ($22\times$ less data, $4.7\times$ worse map noise for fixed survey time), while the contamination *fraction* of what is kept improves by only $1.2\times$ for sub-width signal at $k = 0.1$, $1.6\times$ at $k = 0.3$, and $5.4\times$ even at the width crossing $k = 1$. Below the width crossing — the only regime a deeper cut is supposed to protect — both thresholds retain signal at nearly the null rate, so the purity of kept data barely moves: you cannot tail-cut your way out of $k \ll 1$. What actually bounds that regime is the measurement program (pilot-free floor, injection retention curve, veto arm, per-channel occupancy priors), which pays linearly rather than exponentially.

Four further properties select the mean exactly. (i) It is the one threshold requiring *no width calibration*: $\hat{\mu}_0$ is the best-measured point of the distribution (mode-anchored median, $\sim 3 \times 10^{-5}$ error), while any tail threshold inherits the per-channel width, which carries no assigned uncertainty. (ii) Symmetry makes the null keep-fraction exactly $1/2$, which is what turns the mask fraction into a free, continuous health check — binomial sensitivity is maximal at 0.5 , and a drift test at a 2.3 per cent keep rate would need hundreds of times the frames. (iii) It is parameter-free: $P_d(k) = \Phi(k)$ with no engineered constant — $\Phi(1) = 0.84$ is the measured $0.833/0.841$ — so the whole validation chain anchors to it. (iv) Purity is *not* monotonic in depth in this system: anything that adds power to the reference cells pushes F *down* (measured on ch30’s off-air captures, whose means scatter both ways), so frames far below the mean carry *enhanced* odds of reference-side contamination — the band floor exists precisely because low F is not clean F . The kept set is therefore a window, $[\hat{\mu}_0 - 5\sigma, \hat{\mu}_0]$: the entire trustworthy part of the left tail, cut at the point where every one of these arguments meets.

Recorded options, deliberately not engineered now. Three design notes ratified 2026-07-18, kept for the future rather than built: (1) the per-channel zero-point offsets map to per-channel P_d statements through the Fig.-3 meter — worth an explicit exposition (“how the offset moves P_d ”) when presenting; (2) re-centring a channel’s target weight onto a measured off-nominal carrier (the ch33 option, already recorded in the article) remains a constants-level change within existing machinery; (3) threshold self-calibration: the kernel’s exact num/den readback supports continuous per-channel F -histogram accumulation, so the measured table could regenerate as an automated slow loop with atomic constant swaps (~ 2 M frames/channel/day at 23.8 frames/s — one day of deployment exceeds the whole archive’s per-channel statistics) with the same trust flags refusing contaminated calibrations. All three are recorded to avoid pre-deployment over-engineering, not because they are hard. A fourth, recorded 2026-07-18 from Dylan’s recollection of fid 552: ch33’s occupant may be a *drifting* oscillator (tracked 2019–2023 in prior CHIME monitoring). The per-frame cells already corroborate cell-crossing in the loud era — quarterly low-tail rates of 13–29 per cent through 2018–2020 (reference-cell power depressing F) against a uniform 2.3–5.4 per cent from late 2022, interleaved with the 33–47 per cent detection spikes — reading as an oscillator wandering across the lower reference, guard, and target cells before settling in the guard, where the two accumulated lines would be dwell positions (ch30-picket morphology). If confirmed, this is an alternative mechanism for the 2020 rate collapse (walked out of the cells, vs departed in the repack) and would kill the departed-occupant-ghost reading of the blind spot. A 36-event targeted spectra job (`ch33_drift_events.txt`, `ch33_drift_spectra.py`; one event per quarter, three through the

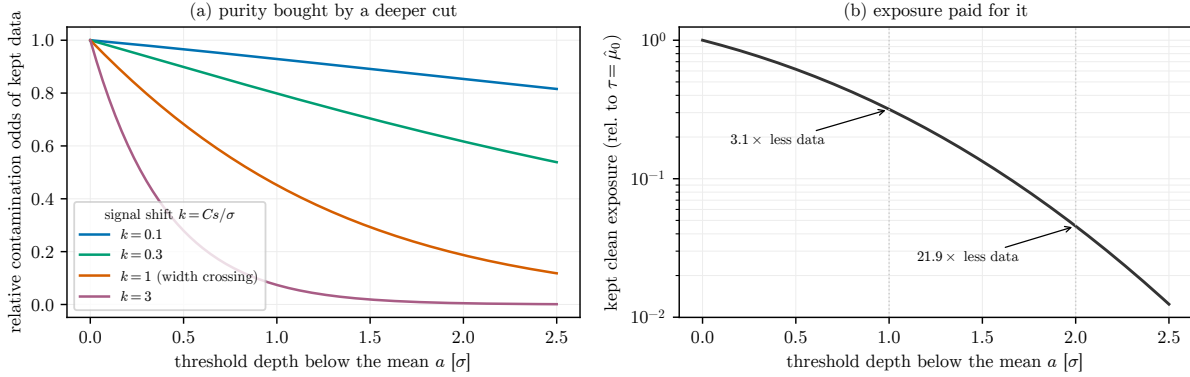


Figure 4: The exchange rate of left-tail thresholds: purity gain per kept frame (a) against clean exposure surrendered (b), for signal shifts k in null widths. The deep tail pays 22× in data for 1.2–1.6× in purity against sub-width signal.

transition, 640 frames, CPU-only, batch-resumable) is prepared and deferred pending the survey extension.

7 Secular transitions, the repack, and a dilution lesson

Seven years of per-frame rates resolve transmitter history directly (`fig_secular_rates`): ch33 collapses ~ 40 to ~ 4 per cent by 2020 Q2 (repack-consistent); ch32 falls ~ 35 to <1 per cent in 2023 April–May (monthly: 0.36, 0.30, 0.09, 0.005); ch35 switches on to ~ 17 per cent in 2021 Q4 — eighteen months after CBUT’s confirmed 2020-05-01 repack move, which is itself a puzzle (interim vs full facility); ch17 rises ~ 5 to ~ 35 per cent across the baseline. All four persist in the classified-FRB stratum (`fig_secular_frb_stratum`), so archive composition does not manufacture them. Two additions came from pushing the scan to *every* calibrated channel: ch34’s step (3–8 per cent through 2020 $\rightarrow \approx 2 \rightarrow \approx 1$) hides below the episodic membership criterion because a full-period tail fraction *dilutes* any epoch that is short against total exposure — a general lesson about full-period selection criteria — and ch27 carries a 38 per cent sampled interval in 2020 Q3 with its interior unsampled. Also worth remembering: the newest quarter (2026 Q2) shows an all-events-elevated / FRB-flat signature on several channels simultaneously — the same compositional artifact as ch33’s spurious 2020 Q4 “rebound.” Rate structure you cannot reproduce inside a uniform trigger class is not transmitter history.

8 ch33: the silent blind spot and the loose-tolerance class

ch33 is the channel every per-frame diagnostic scores as clean — trusted calibration, ≈ 0.5 mask fraction, zero measured leak — and that is precisely the hazard: its only extracted carriers sit in the skipped guard (-3.6 kHz at 21 dB integrated SNR), where the detector’s response is -17 dB. “Clean” and “blind” are indistinguishable from inside the statistic. The taxonomy that came out of this is the risk framing the whole deployment section leans on: a strong co-channel station *on* the nominal pilot is maskable and costs exposure only (ch24, ch30); an *off-nominal* pilot is DTV the mask responds to only weakly and rides into kept data. The census ties the hazard to a licensing class: translator/LPTV entries carry no frequency-tolerance value in the source, and ch33’s census holds 11 such loose-tolerance candidates within 500 mi (`fig1_census_context`). Whether the guard carrier is the current translator or the integrated ghost of the departed pre-repack occupant awaits the epoch-split accumulation — one of the cleanest open questions the dissertation can pose.

9 Instrument tones: marked, justified, and how wide to cut

Of six narrow lines found outside the detector cells across all 23 channels (Fig. 5), five sit within 9.5 Hz — less than half a spectral bin — of $\pm f_s/5$, $\pm f_s/3$, $\pm 2f_s/5$ in baseband ($f_s = 390.625$ kHz). The per-tone case for excision rests on three independent legs, each visible in the gallery: (i) the *arithmetic* coincidence with rational clock fractions, at odds of order 10^{-3} per line by chance; (ii) each line lives in exactly one coarse channel, as chain-specific digital spurs do and as broadcast signals do not; (iii) the prominence is unchanged ($|\Delta| \leq 0.3$ dB) when the analytic mask removes roughly half the frames — a sky signal correlated with DTV presence could not be mask-invariant. The sixth line (ch17, +157.285 kHz) matches no fraction and is deliberately *retained*: unidentified means unexcised. All six sit ≥ 73.6 kHz (24 fine bins) from any detector-cell support, so excision is cosmetic for the statistic either way — notches touch only the supplementary plotting copies.

The notch width itself became a measurement (Dylan’s catch, 2026-07-18): a fixed ± 3 -bin pad left residual skirts of +2.3 dB (ch14), +4.4 dB (ch29, whose feature is 7 bins wide before skirts), and +2.1 dB (ch34) outside the cut, while the narrow $f_s/3$ pair (ch25/28) was clean. The fix is an adaptive notch that grows from the detected group until the spectrum falls within 0.5 dB of the local background (bounded at ± 15 bins, then +3-bin pad), giving extents of ± 8 (ch14), ± 13 (ch29), and ± 9 (ch34) bins — re-audited: no residual above 1 dB anywhere. Lesson worth keeping: spur *skirts* scale with spur strength, so a one-size pad under-cuts exactly the lines that matter most.

The six narrow lines outside the detector cells: five clock-fraction spurs (adaptive notch shown), one retained unknown (mask-invariant prominence $\Delta \leq 0.3$ dB under the analytic mask --- not sky-correlated; nearest line 73.6 kHz from any cell)

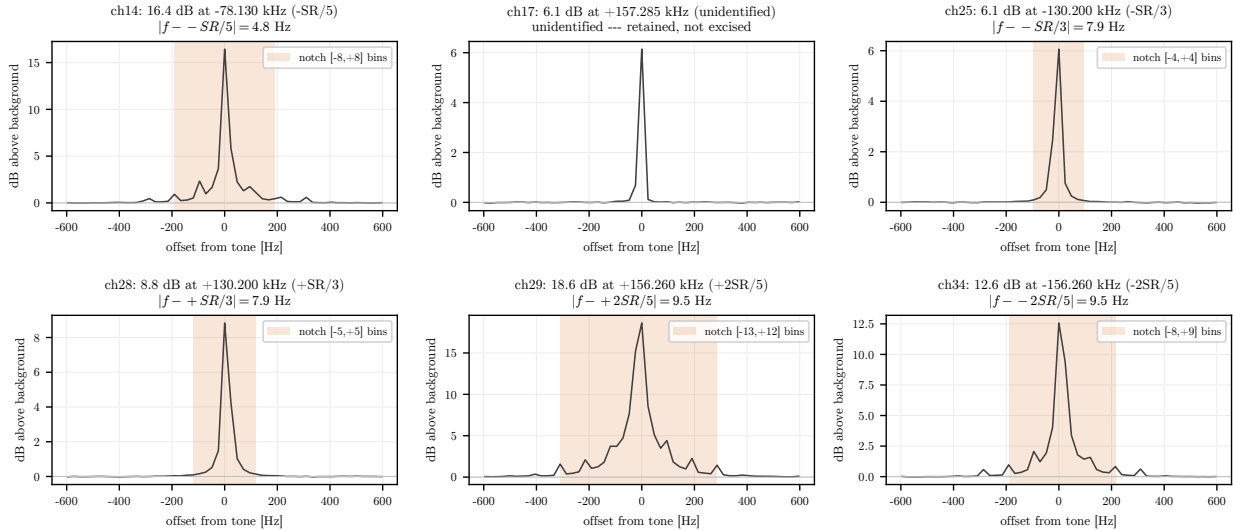


Figure 5: The instrument-tone gallery: every narrow line outside the detector cells, marked and identified, with the adaptive notch extent shaded. The ch17 unknown is retained, not excised.

10 Off-nominal lines vs the census: verified, not assumed

After instrument-tone removal, four channels retain extracted lines outside their target cells. “Rogue transmitters” was the working assumption; the census check makes it a verification. Every one of the four has loose-tolerance candidates licensed on that channel: ch24 (+3121 Hz at 8 dB; 3 relays + 12 translators, nearest K24LM-D at 146 km), ch30 (the drift picket — four lines at $-3899/-3494/+3110/+3611$ Hz, 8–10 dB each, spanning both guards; nearest candidate CHKL-1 itself at 18 km, and four dwell positions of one wandering

carrier integrated over years is the economical reading), ch31 (-2283 Hz at 7 dB ; 3 relays + 17 translators, nearest 79 km), and ch33 ($-3606/-2843\text{ Hz}$ at $21/13\text{ dB}$; 1 relay + 9 translators, nearest K33NM-D at 101 km). No channel shows an off-nominal line with an empty candidate list — the transmitter hypothesis survives everywhere. The ch33 nuance is the honest residual: its nearest loose candidate is 101 km out, so unlike ch30 the attribution is not forced to a single station, and the current-translator-vs-departed-occupant question stays open until the epoch-split accumulation.

11 Survey forensics: joining 77k captures to their events

The archive’s capture units carry only timestamps; the inventory carries only day-level dates and classes. The join that made the stratified analysis possible is ($\text{day} \pm 1$, unit frame count = $\lfloor \text{inventory } n_{\text{frames}} \rfloor$), validated at 97.4 per cent on singleton days; it confidently classes 79.0 per cent of 77,423 units. The naive day-only join is unusable — 89.6 per cent of units fall on mixed-class days. A small method worth writing up: when two datasets share no key, a physical invariant (frame count) can be one.

12 The stack-subset story: recorded procedure vs optimality

The recorded combine rule (signature-seeded exact search) selects $16\text{ channels} \times 1548\text{ common events}$. An unrestricted exhaustive search over all $\binom{23}{16}$ subsets later found a larger block (1829 events) that the seeded procedure *structurally* cannot reach — the seeded candidate table is not even monotone in k . We kept the procedure-as-recorded, measured the robustness across both blocks (stacked headline $3.74\text{ vs }3.53\text{ MHz}$, a composition effect; per-channel and full-depth results invariant), and archived both decision records. The registered greedy rule, for contrast, strands at $22\text{ channels} \times 12\text{ events}$. Preregistration discipline, its limits, and how to report both honestly — dissertation material in itself.

13 Exact-integer thresholds

The measured ceiling and floor ship as per-channel rational pairs (P, Q) with $Q \leq 2^{16}$: approximation error $\leq 10^{-9}$ (four orders below the zero points’ statistical error), zero disagreements against the float rules across all 339,196 archive frames, worst integer product 55 bits. Deployment as a constants swap in an integer compare the kernel already evaluates — the whole calibration apparatus reduced to a table of fractions.

14 The tiering in redshift: depth, not reach

Through $z = \nu_{21}/\nu - 1$, the contaminated band is $1.336 < z < 2.022$ — the high-redshift half of CHIME’s BAO range, capped by the protected $608\text{--}614\text{ MHz}$ allocation. The first-adoption eight gate slices from $z = 1.383$ (ch34) to 1.984 (ch15), summed $\Delta z \approx 0.25$ of the 0.69 available, spread across the interval; every channel the construction cannot yet certify subtracts a $\Delta z \approx 0.02\text{--}0.03$ notch (ch24: $1.650\text{--}1.680$; ch30: $1.483\text{--}1.510$; ch33: $1.407\text{--}1.432$) rather than a contiguous range. The conservative tiering costs depth on slices, never redshift reach — the one-sentence answer to “what does this buy the cosmology.”

A Plot annex: the full supplementary set

The complete diagnostic set at full size — the three spectra variants over every coarse channel (raw; instrument tones + DC notched; after the analytic positive-excess mask), the pilot-zoom view that validates the cell geometry on-sky, the per-channel F distributions about the operating threshold, and the diurnal mask-fraction diagnostic. PNG twins of every panel live in `supplemental_plots_rev16a.zip`.

Integrated spectra, full coarse channel (dc-centred), all 23 channels (mean per valid frame, dB rel. channel median; detector cells shaded) --- before mask (raw)

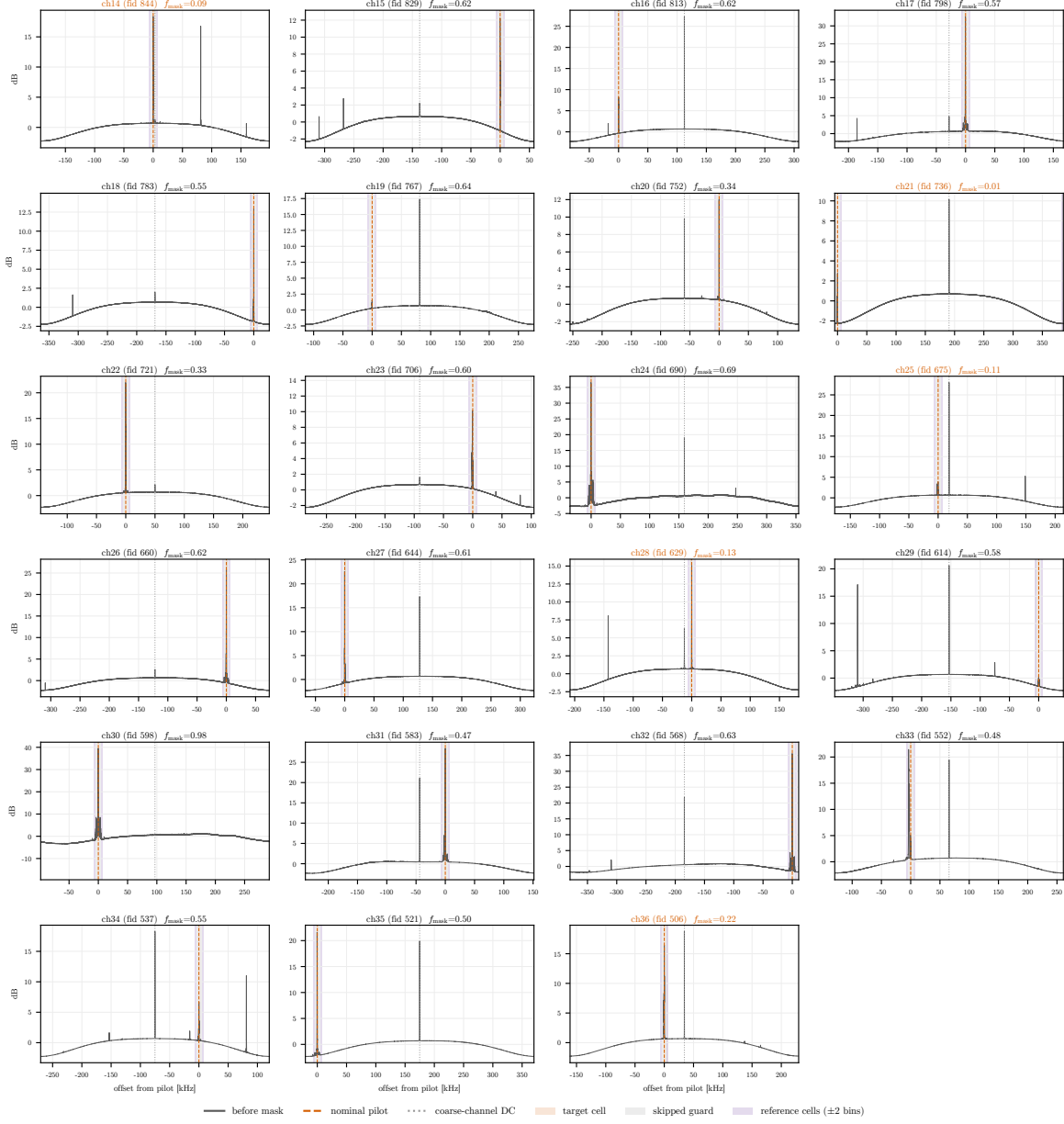


Figure 6: All 23 coarse channels, integrated spectra, raw (before any notch or mask). Detector target/guard/reference cells shaded.

Integrated spectra, full coarse channel (dc-centred), all 23 channels (mean per valid frame, dB rel. channel median; detector cells shaded) --- before mask, instrumental tones + DC notched to local background (plotting copies only)

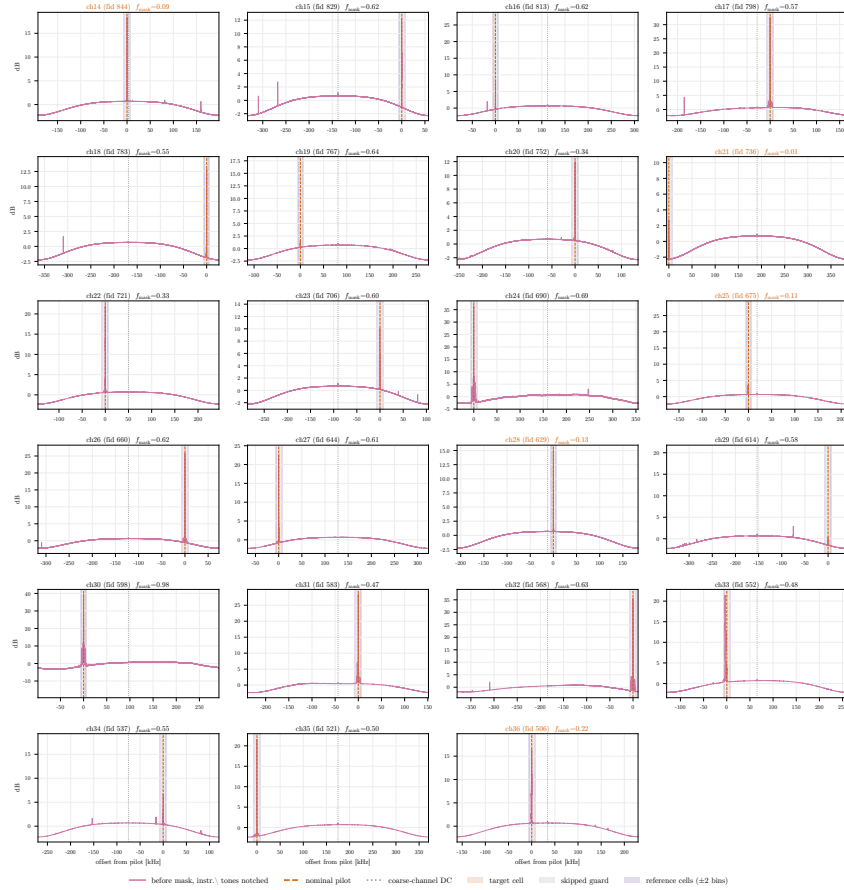


Figure 7: Same, with the coarse-channel DC spur and the five identified clock-fraction tones notched to the local background (adaptive extents, §9).

Integrated spectra, full coarse channel (dc-centred), all 23 channels (mean per valid frame, dB rel. channel median; detector cells shaded) — after the original analytic positive-excess mask (not the candidate three-arm rule), instrumental tones + DC notched

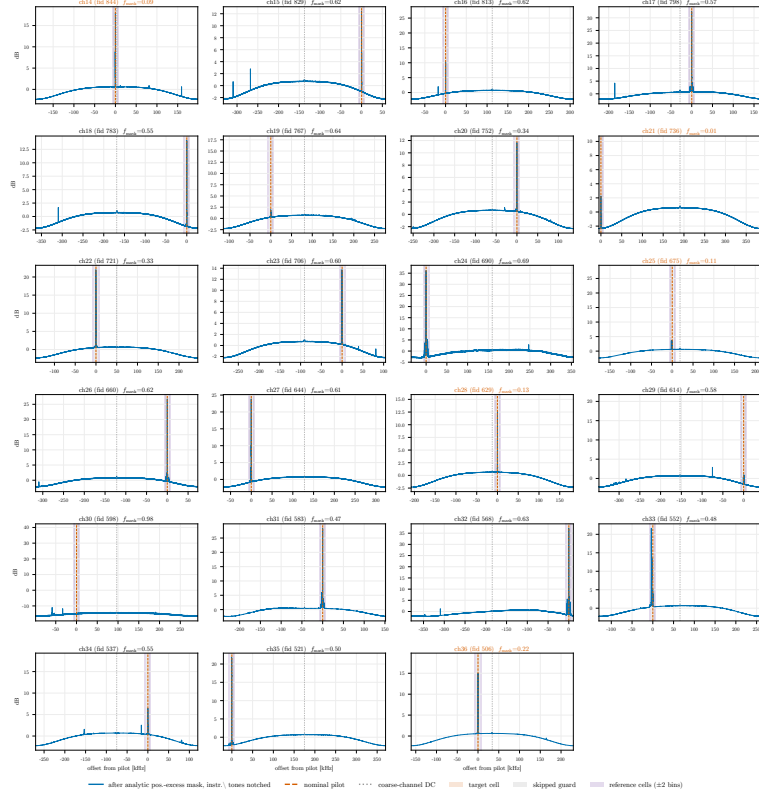


Figure 8: Same, after the analytic positive-excess production mask — what the detector actually removes.

Integrated spectra, ± 20 kHz about the nominal pilot (target / skipped-guard / reference cells shaded, one fine bin wide; extracted spectral lines as ticks) --- before mask, instrumental tones + DC notched to local background (plotting copies only)

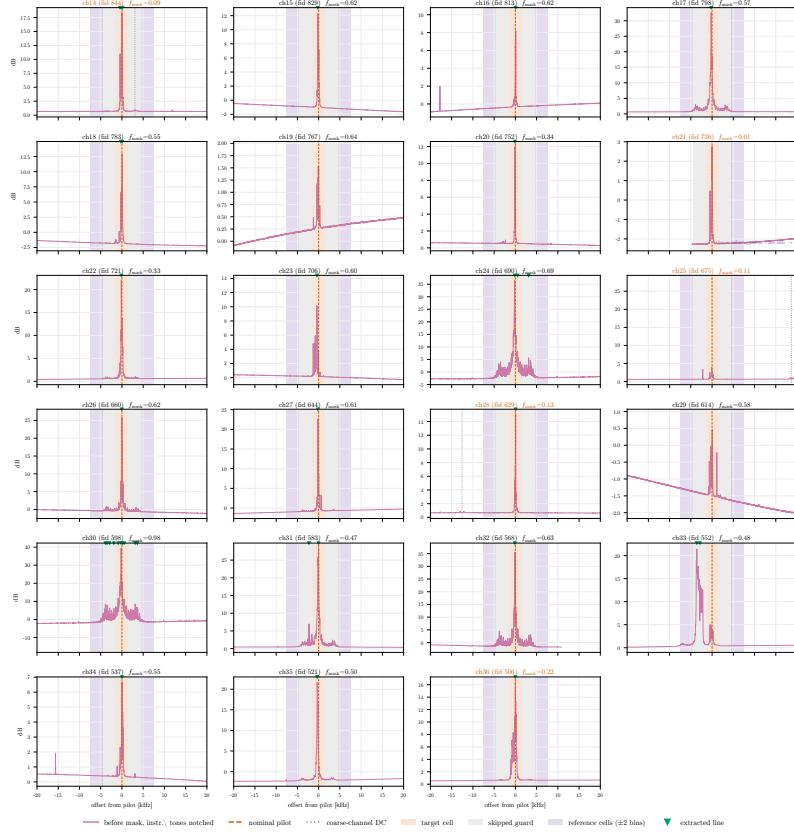


Figure 9: Pilot-zoom, instrument-removed: the on-sky cell-geometry validation view (extracted carriers vs target/guard/reference cells).

Valid-frame F distributions about the operating threshold, all 23 channels (kept vs masked; sub-threshold signal leak from mirrored-core estimate)

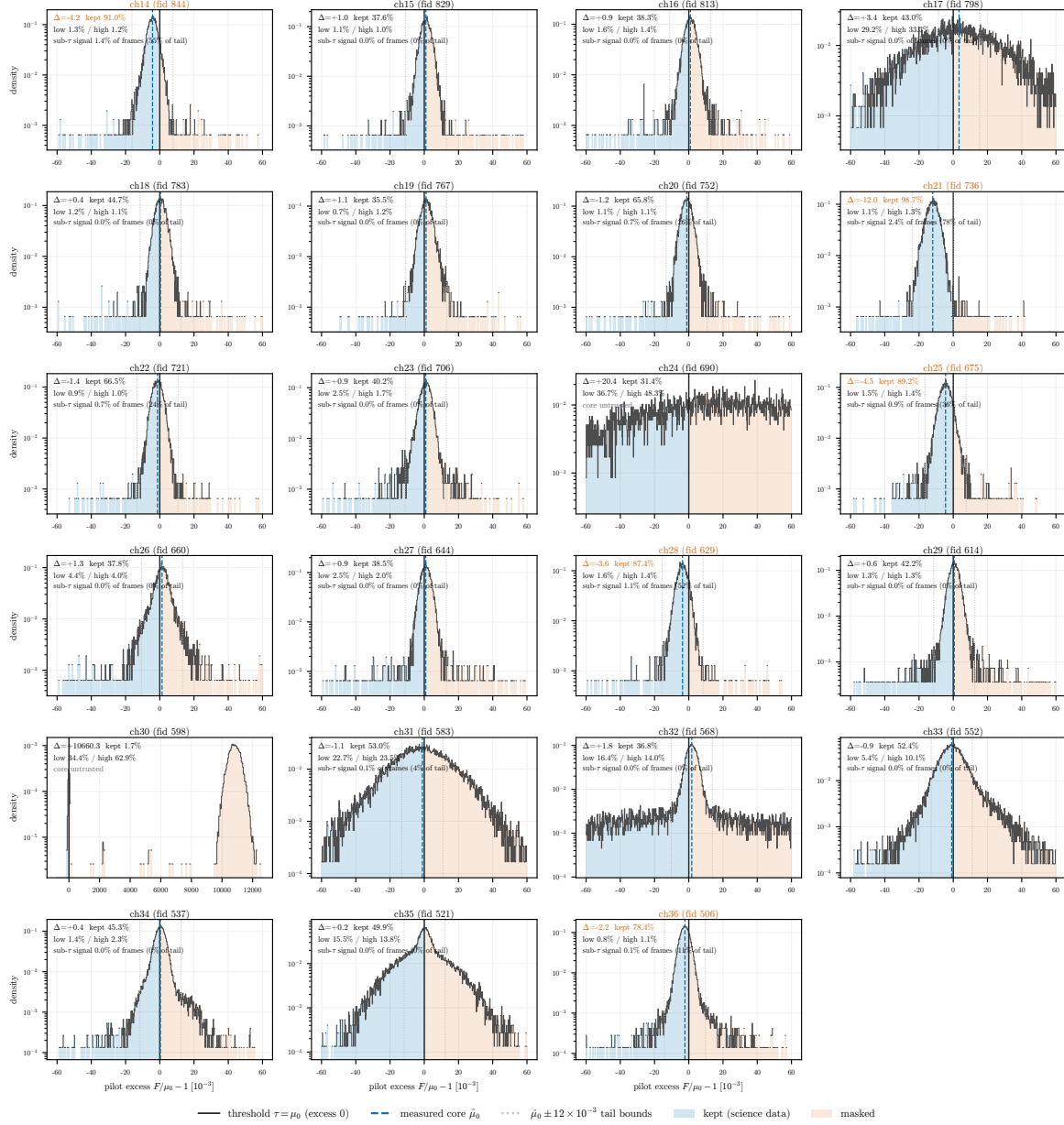


Figure 10: Per-channel F distributions about the operating threshold: kept/masked split, both zero points, tail bounds, and the sub-threshold leak estimate in a single view.

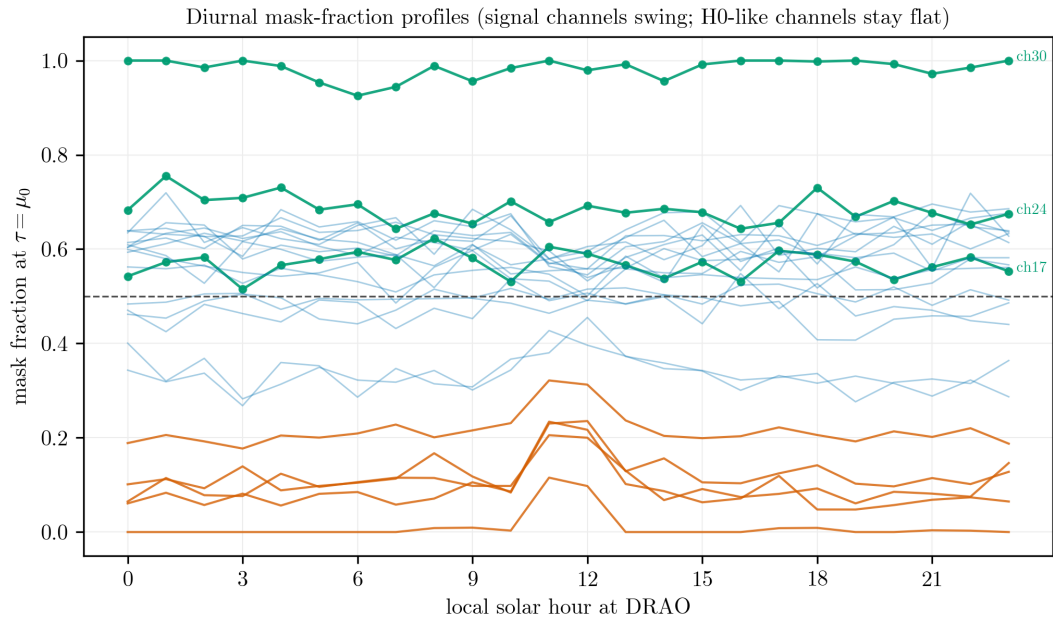


Figure 11: Diurnal mask-fraction diagnostic.

Living-document note. Sources for every section: the public repository (WVURAIL/pilot-proxy), the archived provenance memos (DECISIONS, T2_RESOLUTION, SURVEY_COMPOSITION_STRATUM, TIERING_COMPLETENESS, referee-triage series R5–R10), and the per-figure scripts named in each section. New nuggets get appended as they are found; entries are dated in the git history of this file.